Parametric pump

An LC tank whose capacitor is modulated at twice resonance: envelope growth against Mathieu's rate, with the pump's work on the ledger — CLAUDE.md §3a, M9c.1

kronspice simulation report · 2026-11-07

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Summary

A one-netlist example of kronspice's time-varying capacitor (the SCALE= extension). A 1 mH, 1 µF tank (f_0 = 5.033 kHz) starts with 1 V on the capacitor, and the capacitance is driven as C(t) = 1 µF·(1 − 0.1·sin $2\omega_0 t$): it falls while |v| peaks, so the tank's energy grows. It is the capacitive twin of the pumped inductance in examples/kron_machine, and the simplest form of the parametric circuits of H16 (examples/parametric_lcr). It gives the standard prediction beside which H16's "energy from the time domain" is read: the agent that varies C pays.

- **Envelope growth:** v(a) grows from 1.02 V at the first peak to 4.44 V at 1.992 ms, an exponential rate of 740.8 /s.
- **Mathieu's rate** for pump depth m at $2\omega_0$, less the series damping, is $m\cdot\omega_0/4 - R/(2L)$ = 790.6 − 50 = 740.6 /s (Kouřil & Vrba §10.2.2, eq. 10.101).
- **The pump pays:** net pump work 10.36 µJ (11.70 µJ in through the mechanical port, 1.33 µJ returned) equals the stored-energy gain (0.50 → 10.25 µJ) plus the heat in R1 (0.61 µJ).
- **Ledger:** integration mismatch 1.2 nJ over the 2 ms; worst point residual 0.

The netlist

```
C1 a 0 1u SCALE=SIN(1 0.1 10.066k 0 0 180) IC=1
L1 a b 1m
R1 b 0 0.1
.tran 1u 2m UIC
```

SCALE=wave(...) multiplies the element's charge curve by s(t): the charge is q = s(t)·C·v, so the current is i = C·s·dv/dt + C·v·ds/dt. Here s(t) = 1 − 0.1·sin($2\omega_0 t$), at 10.066 kHz = $2f_0$, with a 180° phase. The work of changing s, ṡ·∫Q, is the work of whatever agent squeezes the capacitor, and it is booked on the element's **mechanical port** (an EXT= channel can redirect it).

The 180° phase makes s fall as |v| peaks, so the pump puts energy in. With 0° the tank decays and the port takes energy out instead (analysis.TestParametricPumping).

Result

quantity	value
envelope of v(a), first peak	1.02 V
envelope of v(a) at 1.992 ms	4.44 V
measured exponential rate	740.8 /s
Mathieu first-order rate $m\cdot\omega_0/4$	790.6 /s
series damping R/(2L)	50 /s
predicted net rate	740.6 /s

analysis.TestParametricSelfExcitation holds the engine to this rate.

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Energy ledger over the 2 ms

entry	energy
in through the mechanical port	11.70 µJ
returned to the port	1.33 µJ
dissipated in R1	0.61 µJ
stored, start → end	0.50 µJ → 10.25 µJ
integration mismatch	1.2 nJ

The net work of the pump, 10.36 µJ, equals the stored-energy gain plus the heat: nothing comes from nowhere, and the ledger's worst point residual is 0. This is the standard prediction beside which H16 is read (CLAUDE.md §1): the energy the tank gains enters through the capacitor's mechanical port, from the agent that varies C.

Validation

- analysis.TestParametricSelfExcitation: envelope growth rate against Mathieu's first-order rate.
- analysis.TestParametricPumping: 180° phase pumps energy in; 0° phase decays, with the port taking energy out.
- export/ holds the netlist, a KiCad schematic and the operating-point sheet (MNA only: Kron's torn solve on the sheet is linear-only, and C1 is a nonlinear capacitor).

Limits

- The modulation s(t) is imposed; the agent that varies C is ideal and appears only through its work on the port.
- The growth rate is compared with Mathieu's first-order rate (pump depth 0.1).

Reproducing the results

```
go run ./cmd/kronspice examples/parametric_pump/pump.cir
go run ./cmd/kronspice -o pump.csv examples/parametric_pump/pump.cir     # waveforms and ledger columns
go test ./analysis -run 'TestParametricPumping|TestParametricSelfExcitation'
```

Sources: F. Kouřil, K. Vrba, *Non-linear and Parametric Circuits*, Ellis Horwood 1988, §10.2.2; L. Mandelstam, N. Papalexi, *Z. Phys.* 73, 223, 1932; doc/M9c1.md (the pumped inductance against Mathieu's rate).

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